

This is an estimate, not a measurement. No slicer is available in this environment to compute real filament usage, and none of the STL files are watertight solids (so exact volume can't be read directly either). Numbers below come from each part's real mesh *surface area* multiplied by an effective wall/infill thickness derived from the confirmed print settings – a reasonable approximation, but not a substitute for an actual slice. Treat every figure here as a planning number for how much filament to have on hand, with healthy margin, not a precise bill.

Method & assumptions

- Each part's real surface area (from its current STL) × an effective shell thickness = approximate printed volume.
Effective thickness = (wall count × 0.42mm line width) + (infill % × 0.42mm), using 0.42mm as a typical line width for a 0.4mm nozzle.
- This approximates a part as "shell + thin infill core," which holds up well for the drum and container (large, relatively thin-walled shells) – it's less exact for small chunky parts, but close enough for a material-planning estimate.
- PETG density: 1.27 g/cm³.
- Default print setting (confirmed in build-notes.md): 3 walls, 5% infill – used unless a part has its own confirmed override (slip-in inserts: 4 walls/30% gyroid; bristle-guard foot: 4 walls/30% gyroid).
- Overhead multipliers added on top of the core material figure:** +8% for brims (most parts in this build use some brim for adhesion) and tree supports (overhangs on the container's dome, spindle bracket area, and similar features), plus a further +10% general waste/reprint buffer – this build has already gone through many reprints of the container and inserts during development, so a buffer for a bad print or two is realistic, not pessimistic.
- Parts marked measured use the actual current STL file's real surface area. Parts marked sized guess have no STL on hand in this session (electronics lids, pulley, drawer assembly, retainer cap, spindle shims) – their mass is a rough guess based on comparable part sizes elsewhere in this build, not a geometry measurement. These are the least reliable numbers on this page.

1. Drum & container (the two big prints)

Part	File	Surface area	Settings	Core mass	+19% overhead
Drum measured	Drum_LATEST_chamfered_bristleholes_v3_final.stl	1,986,736 mm²	3 walls / 5% infill	3.23 kg	3.84 kg
Container measured	AutoPooper_Container_v14_symmetric.stl	3,444,371 mm²	3 walls / 5% infill	5.60 kg	6.66 kg
Subtotal – drum + container, with overhead					10.50 kg

These two parts dominate the material budget by a wide margin – together they're roughly 85% of the whole project's PETG, before secondary parts.

2. Secondary PETG parts – measured from current STLs

Part	Qty	Settings	Mass (ea.)	Mass (total)
Shaft-holder slip-in insert, L measured	1	4 walls / 30% gyroid	45 g	45 g
Shaft-holder slip-in insert, R measured	1	4 walls / 30% gyroid	47 g	47 g
Height shims (L/R × 1mm/2mm, all 4) measured	4	3 walls / 5%	~11 g	44 g
Heat-set reinforcement bushing measured	8	3 walls / 5%	1.5 g	12 g
Nameplate measured	1	3 walls / 5%	170 g	170 g
Spindle holder cap (v26) measured	1	3 walls / 5%	111 g	111 g
Upper spindle bracket (one variant installed) measured	1	3 walls / 5%	189 g	189 g
Subtotal – measured secondary parts				618 g

(Pilot-hole reference disc excluded – it's a check tool, not an installed/printed part.)

3. Secondary PETG parts – no STL on hand this session, sized by comparison

Part	Qty	Guessed mass range
Electronics bay lid sized guess	1	150 – 250 g
Access panel lid sized guess	1	100 – 150 g
Pocket access plug sized guess	1	20 – 40 g
Belt pulley (HTD 40-8M-20) sized guess	1	80 – 150 g
Retainer cap sized guess	1	30 – 60 g
Drawer (one-piece) sized guess	1	300 – 600 g
Drawer tray sized guess	1	150 – 300 g
Drawer faceplate sized guess	1	80 – 150 g
Drawer handle sized guess	1	30 – 60 g
Drawer lid sized guess	1	100 – 200 g
Spindle bracket & leg shims (all 6 thicknesses) sized guess	6	60 – 120 g
Subtotal – unmeasured secondary parts		1,100 – 2,080 g

These numbers are rough – call them "have this much on hand," not "this is exactly what it'll use." The drawer assembly is the single biggest unknown here since it's the largest unmeasured part.

Overall project total (PETG)

Category	With overhead
Drum + container	10.50 kg
Measured secondary parts (+8% overhead)	0.67 kg
Unmeasured secondary parts, midpoint guess (+10% overhead)	~1.65 kg
Running total before final waste buffer	~12.8 kg
+10% general reprint/waste buffer (this build has gone through many iterations already)	~1.3 kg
SPECULATIVE PETG TOTAL – WHOLE PROJECT	~12.5 – 15.5 kg

Practical takeaway: budget for roughly **13-16 kg of PETG** (a standard 1kg spool run, so 13-16 spools) to comfortably cover the whole build including reprints, with the drum and container alone eating about 10.5 kg of that. This does **not** include the dump-chute/poop-ramp part (not designed yet) or the ACF (carbon filter) housing (also not designed yet) – add both once those exist.